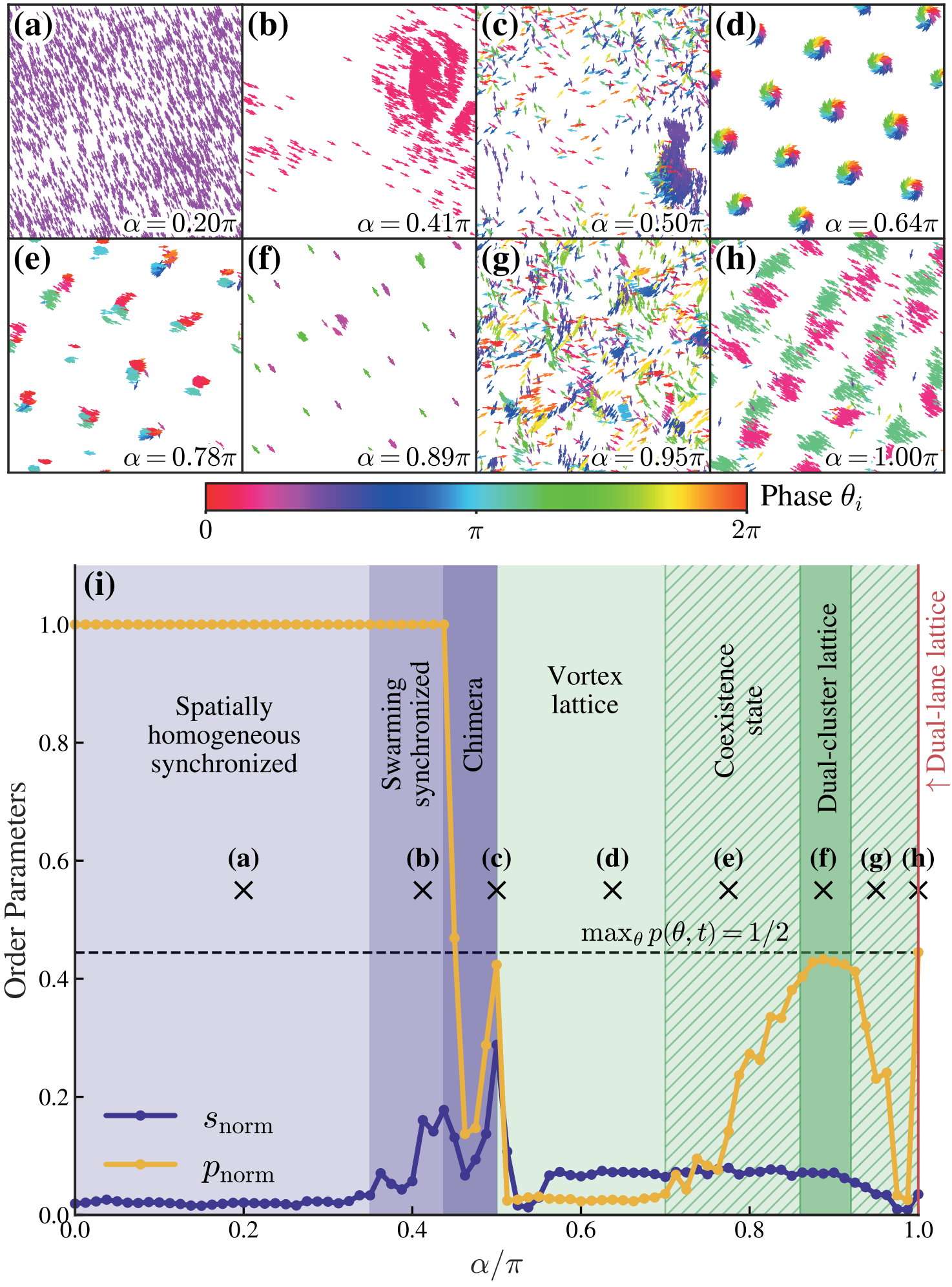
(a)
$\alpha = 0.20\pi$
(b)
$\alpha = 0.41\pi$
(c)
$\alpha = 0.50\pi$
(d)
$\alpha = 0.64\pi$
(e)
$\alpha = 0.78\pi$
(f)
$\alpha = 0.89\pi$
(g)
$\alpha = 0.95\pi$
(h)
$\alpha = 1.00\pi$
Phase θ_i
0
π
2π
(i)
Spatially homogeneous synchronized
Swarming synchronized
Chimera
Vortex lattice
Coexistence state
Dual-cluster lattice
↑Dual-lane lattice
$\max_\theta p(\theta, t) = 1/2$
s_{norm}
p_{norm}
Order Parameters
α/π
0.0
0.2
0.4
0.6
0.8
1.0